Examiner reports

Q1.

Part (a) was completed successfully by most students. A significant minority used $\frac{\theta}{2\pi} \times \pi r^2$ when finding the area of the sector, possibly demonstrating a lack of practice with the standard formulae.

Part (b) tested knowledge of the Newton-Raphson method, which is new to the A-level maths specification. Less than half of students were successful on this question, with many not realising that the method is used to solve an equation of the form $\mathrm{f}(x) = 0$. Students who did rearrange the given equation often scored full marks.

Q2.

Most students began part (a)(i) successfully forming an equation with $V = 0$ to achieve the first mark. Having started well, only around half of these students went on to demonstrate the algebraic maturity required to arrive at the given result.

Part (a)(ii) was completed successfully by many students. The majority scored fully marks.

In part (a)(iii) relatively few students identified that $T_0 = 38$ represented the current year. A significant number of students made no attempt at this question.

The most frequent mark for part (b) was 2/4, with many students starting well and evaluating **both** models at $t = 49$ or $t = 50$ and indicating that they were close together. Few students went on to correctly complete a change of sign argument, which would have shown the required result.

An alternative method was to form an equation from the two models and solve numerically using allowed calculator functions. This method was accepted as fully correct provided it was explained thoroughly, as shown in the mark scheme.

Q4.

Part (a) was well answered by the majority of the students. Most students used $\mathrm{f}(x) = x^3 - 6x + 1$ and evaluated f(2) and f(3) correctly. There were students who then wrote "change of sign therefore a root" without clarifying where the root lies. Very few students rearranged the expression to use the alternative LHS / RHS method; and those who did so were less successful as they appeared to be unable to make a correct statement with many still just stating "change of sign therefore a root".

Part (b) was generally correctly derived with students occasionally losing the mark through the omission of " = 0".

Part (c) was answered very well but a minority of the students failed to round off their answer to four significant figures.

Q5.

The great majority of students gained full marks for this question.

(a) The evaluations of the function for x = 3 and x = 4 were correct in almost all scripts. But some students failed to make a full statement that mentioned both the change in sign of the function as well as the presence of a root between x = 3 and x = 4.

(b) Almost all students gained full marks. A few students confused “three decimal places” with “three significant” figures or gave 3.88 instead of 3.880 for x_2 .

(c) Almost all students gained both marks except for very few cases where x_2 and x_3 were not marked on the x -axis.

Q6.

The majority of students scored full marks for this opening question which tested the Newton-Raphson method. The common errors were mainly arithmetical, with an incorrect evaluation of f' (10) as 324, instead of its correct evaluation as 284, being the most common. A small minority of students did not give their answer to the required degree of accuracy and so did not score the final mark. This question was not attempted by significantly more students than usual. Not all non-attempts were from the weaker students. It should be noted that the required formula is given in the formulae booklet.

Q7.

Part (a) was omitted by a significant minority of candidates but even some of these candidates gained some credit for applying Newton-Raphson method in part (b). The most common errors in part (b) were to either take $f(x_n)$ as 2 instead of –10 or to leave the final answer as $\frac{13}{4}$ instead of giving the exact decimal 3.25.

Q8.

This was well done with many students obtaining full marks.

In part (a), the majority of students used $4\ln x - \sqrt{x}$ = 0 and evaluated it correctly for x = 0.5 and

x = 1.5. There are still many students who then write ‘change of sign therefore a root’ without clarification of where the root lies and who therefore lose the A mark. Less successful were the students who tried the LHS-RHS approach on $4\ln x = \sqrt{x}$. Although correct numerical values were seen earning the method mark, the final accuracy mark was often not earned.

Parts (b) and (c) were well answered by the majority of students. When students failed to score full marks, it was usually due to poor notation.

Part (d) was well answered by the majority of students. The main error was the incorrect labelling of the axes.

Q9.

This question, which tested the numerical methods section of the specification, was another good source of marks for the students. In part (a), most students correctly evaluated the given cubic expression (= $f(x)$) at both x = 0.1 and x = 0.2 and stated 'change of sign' but some then stated incorrectly that the root lies between –2.816 and 0.232, or failed to state the values between which the root lies, and so did not score the final mark. In part (b), many students applied the Newton-Raphson method correctly and scored full marks, although some other solutions were presented which contained arithmetic errors, or a final answer which was not rounded to the four decimal places required.

Q10.

Part (a) was well answered by the majority of candidates. Candidates should be encouraged to write down their relevant function before they begin to substitute – too many fudged their approach.

Most candidates used $f(x) = (x^2 - 4)\ln(x + 2) - 15$ and evaluated f(3.5) and f(3.6) correctly. There are still many candidates who still then write 'change of sign' therefore a root without clarification of where the root lies. Those candidates who used the alternative $f(x) = (x^2 - 4)\ln(x + 2)$ and then compared it with $y = 15$ were less successful, as they appeared to be unable to then make a correct statement.

Part (b) was very well answered, the only real errors being ± missing in the final answer and some candidates losing a bracket in their working.

Part (c) was very well answered with full marks often obtained. The main error was stating x_3= 3.567 through using premature approximation of x_2.

Q11.

In part (a) was good to see that a much greater proportion of the candidates defined $f(x)$ first before they substituted values. As it is essential to get the correct numerical outcomes here it is well worth checking before moving on (eg quite a few carelessly rounded 0.0779 to 0.8). Those who used degrees could make no progress. Many lost the second mark by failing to state their conclusion and that α was in the required interval. Those who found the four numerical values needed to be precise in their statements to earn the accuracy mark.

Part (b) was well done.

Part (c) was well done with almost everyone giving both answers to the required accuracy, but those whose calculator was in degree mode earned zero.

Q12.

In part (a)(i), even though the sketch was often poorly drawn, many candidates obtained full marks, and, where they did not, it was often because sketches went beyond the correct end points. Some correct graphs had the wrong end points marked, reversed coordinates being the most common error. There were however many candidates who had no idea what the graph looked like.

Not all candidates attempted part (a)(ii). Where lines were drawn, they were often not accompanied by sufficient explanation to award the accuracy mark. There were also many instances of lines with a positive gradient intersecting in the third quadrant.

In part (b), where candidates clearly defined which function they were using many achieved full marks. There are still candidates who lose marks by saying "change of sign so the root lies between these two values" without stipulating $0.5 < x < 1$.

Part (c)(i) was usually well answered, with answers given to the required degree of accuracy.

Part (c)(ii) was very well answered with most candidates obtaining both marks.

Q13.

(a) This was well answered by the majority of candidates. Many fully correct responses were seen. Most candidates used $f(x) = 3^x - 10 + x^3$ and evaluated f(1) and f(2) correctly. There are still many candidates who still then write 'change of sign' therefore a root without clarification of where the root lies. Those candidates who used the alternative LHS RHS method were less successful, as they appeared to be unable to then make a correct statement, with many just putting 'change of sign' therefore a root.

(b) (i) This part was very well answered, the only real error being 3^x becoming $3x$ during rearrangements.

(ii) Again this part was very well answered. Many fully correct responses were seen. The main error was with candidates who wrote answers to 3 significant figures rather than 3 decimal places.

Q14.

Part (a) of the question was reasonably answered with most candidates obtaining the method mark for substitution of the two given values. Some candidates then lost the accuracy mark through inaccurate evaluation, although the main reason for the loss of the final mark was because many candidates just put 'change of sign therefore root between' without stipulating 0.5 < root < 1.

Part (b) was answered very well with most candidates obtaining the mark.

Most candidates were able to do part (c), with the correct answers often seen.

In part (d), the cobweb diagram was very well done, with the majority of candidates obtaining both marks. Some candidates lost the final mark through inaccurate positioning of x_2 and x_3 on the actual graph and not on the horizontal axis.

Q15.

Part (a)(i) was not very well answered by the majority of candidates. Fully correct responses were seen but it was usually from candidates who successfully rearranged the equation into the form $f(x) = 0$, which was seen in a number of acceptable forms. Where candidates simply substituted in the two values given into the LHS they obtained 1 and 0 but still indicated that this was a change of sign and therefore there was a root. Most candidates achieved at least one mark but lost the second by simply stating change of sign therefore a root.

Very few incorrect responses were seen in part (a)(ii).

Part (a)(iii) was very well answered. The main error was with the candidates who used degrees rather than radians

Part (b)(i) was very well answered, with most candidates successfully using the quotient rule. Where errors occurred it was usually through missing brackets, although many candidates were able to recover at some stage in the working.

As part (b)(ii) followed on from part (i) most candidates were able to obtain the mark for substitution; even those who had incorrectly simplified their work. Many fully correct responses were seen. Several candidates also stopped after the substitution of x = 0 and left their answer as –2.

Q16.

Part (a) of this question was reasonably well answered with most candidates obtaining the method mark for substitution of the two given values. Some candidates then lost the accuracy mark through inaccurate evaluation, although the main reason for the loss of the final mark was that many candidates just stated "change of sign therefore root" without stipulating that $-0.33 < \alpha < -0.32$.

Part (b) was answered very badly. $x^4 + (1 + 3x) = 0$ was a common error as was $(1 + 3x)^{1/4} = -x$, which then became $(1 + 3x) = -x^4$. This part was omitted on many scripts.

Most candidates were able to score full marks on part (c), with the correct answer often seen. One error was to write the final answer to two decimal places as –0.33. The other main error was calculating -0.33^4 and not $(-0.33)^4$ for x_n^4 in the numerator.

Q17.

Most candidates performed well in this question.

Part (a)(i) seemed to present a stiff challenge to their algebraic skill, though they often came through the challenge successfully after several lines of working. Part (a)(ii) again proved harder than the examiners had intended, some candidates having to struggle to evaluate the y-coordinate of the point A. Again the outcome was usually successful, though a substantial minority of candidates used differentiation of the answer to the previous part. In part (a)(iii) it was pleasing to see that a very good proportion of the candidates correctly mentioned h 'tending to' zero rather than 'being equal to' zero, which was not allowed, though the correct value of the gradient could be obtained by this method and one mark was awarded for this.

The responses to part (b) suggested that most candidates were familiar with the Newton-Raphson method and were able to apply it correctly, but that they lacked an understanding of the geometry underlying the method, so that they failed to draw a tangent at the point A on the insert as required, or failed to indicate correctly the relationship between this tangent and the x-axis.

Q18.

Part (a) had a mixed response. Quite often answers were expressed as decimals and equally frequently in degrees, with and A = 180 and B – 90 being common errors. Part (b) was a good source of marks for many candidates. Those candidates who correctly arranged the equation as $\cos^{-1} x - 3x - 1 = 0$ were usually successful in answering this question. Problems arose for candidates who tried to rearrange to other forms such as x = cos ($3x$ + 1), and then were unable to explain their results. Part (c) was usually very well answered, with most candidates receiving full marks.

Q19.

This question was probably the best answered on the paper, with many fully correct solutions.

The only section where candidates lost marks was part (c)(i) due to incorrect calculations or rounding.

Q20.

Part (a) This was well answered by the majority of candidates. Many fully correct responses were seen and if there were errors it was usually in the conclusion.

Part (b) Very few incorrect responses seen.

Part(c) Very well answered. Some candidates lost the final mark since they did not write their answer to 3sf. Other errors involved the use of an incorrect iteration $\sqrt[3]{x_n + 7}$.

Q21.

Most candidates were able to differentiate f(x) correctly and to apply the Newton-Raphson method correctly in parts (a) and (d). Part (b) tested their understanding of the method and was not nearly as well done. Part (c) was only rarely done properly; many candidates using the magnitudes of the values of f(x) rather than their signs to supply evidence for the proposition put before them.

Q22.

This question was the least well answered question on the paper with very few correct solutions. Although the graphs of inverse trigonometrical functions is stated in the specification there were many incorrect graphs sketched. Many candidates had their calculators in degree mode and in part (b)(ii) found the values of 19 and 21 and then stated, ‘change of sign, therefore root’. There were similar mistakes in part (c) with iterations jumping from 0.8 to 44 with candidates apparently not realising there was something amiss.